NOTES (OCTOBER 20)

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Made from handwritten notes by myself with chatGPT and editing.

1. Review of last time

Let $(A, \mathfrak{m}, k)$ be a Noetherian complete local ring and let E be the injective hull of k, which is characterized (see Prop 2.2 below) by the following properties.

- E is an injective module;
- $\mathrm{Hom}_A(k, E) = k$;
- $E \in \mathrm{D}_{\mathrm{tors}}(A)$.

Then we have the normalized dualizing complex

$$\omega_A^\bullet = \widehat{E}.$$

We also have

$$\mathrm{RHom}_A(k, \omega_A^\bullet) \simeq k, \qquad \mathbb{D}(M) = (\mathrm{R}\,\Gamma_{\mathfrak{m}} M)^\vee,$$

where $\mathbb{D}(M) = \mathrm{RHom}_A(M, \omega_A^\bullet)$ and $X^\vee = \mathrm{RHom}_A(X, E)$.

We have

$$\mathbb{D}(M) \in \begin{cases} \mathrm{D}^{[-b-\dim A, -a]}(A) & \text{for } M \in \mathrm{D}^{[a,b]}(A), \\ \mathrm{D}_{\mathrm{coh}}(A) & \text{for } M \in \mathrm{D}_{\mathrm{coh}}(A), \end{cases}$$

and the biduality isomorphism

$$M = \mathbb{D}(\mathbb{D}(M)) \quad \text{for } M \in \mathrm{D}_{\mathrm{coh}}(A).$$

2. Uniqueness

Lemma 2.1. *Let $X \in \mathrm{D}^+_{\mathrm{tors}}(A)$. If*

$$\mathrm{RHom}_A(k, X) \in \mathrm{D}^{\geq \alpha}(A),$$

then $X \in \mathrm{D}^{\geq \alpha}(A)$.

Proof. Let β be such that $X \in \mathrm{D}^{\geq \beta}$ and $X \notin \mathrm{D}^{\geq \beta+1}$. Then

$$\mathrm{H}^\beta \mathrm{RHom}_A(k, X) \simeq \mathrm{Hom}_A(k, \mathrm{H}^\beta(X)) \neq 0.$$

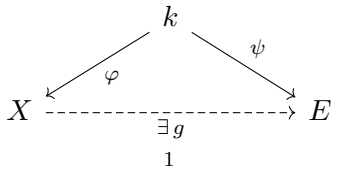
Thus $\beta \geq \alpha$. □

Proposition 2.2. *Let $X \in \mathrm{D}^+_{\mathrm{tors}}(A)$ be such that $\mathrm{RHom}_A(k, X) \simeq k$. Then $X \simeq E$.*

Proof. The proof proceeds as follows.

(1) $X \in \mathrm{D}^{\geq 0}$ (Lemma 2.1).

(2) Let $\varphi : k \to X$ be a generator of $\mathrm{Hom}_{\mathrm{D}(A)}(k, X) = \mathrm{H}^0(\mathrm{RHom}_A(k, X)) \simeq k$, and $\psi : k \to E$ be a generator of $\mathrm{Hom}_{\mathrm{D}(A)}(k, E) = \mathrm{H}^0(\mathrm{RHom}_A(k, E)) \simeq k$, Set $C = \mathrm{cofib}(\varphi)$. Then $C \in \mathrm{D}^{\geq 0}(A)$, so so $\mathrm{RHom}_A(C, E) \in \mathrm{D}^{\leq 0}(A)$, therefore in the following diagram g exists.



(3) By choice, both φ and ψ induce an isomorphism on $\mathrm{Hom}_{\mathrm{D}(A)}(k,-)$, and thus by assumption, g induces an isomorphism on $\mathrm{RHom}_A(k,-)$. Hence

$$\mathrm{RHom}_A(k,\mathrm{cofib}(g))=0,$$

which implies $\mathrm{cofib}(g)=0$ (Lemma 2.1).

Consequently $X\simeq E$. □

Proposition 2.3. *Let $Y\in \mathrm{D}^+_{\mathrm{comp}}(A)$ be such that $\mathrm{RHom}_A(k,Y)\simeq k$. Then $Y\simeq \omega_A^\bullet$.*

Proof. We have

$$\mathrm{RHom}_A(k,\mathrm{R}\,\Gamma_{\mathfrak{m}}Y)\simeq k,$$

so $\mathrm{R}\,\Gamma_{\mathfrak{m}}Y\simeq E$ by the preceding proposition. Hence

$$Y\simeq \widehat{\mathrm{R}\,\Gamma_{\mathfrak{m}}Y}\simeq \widehat{E}=\omega_A^\bullet.$$

□

3. Dualizing complexes: the standard definition

Definition 3.1. Let A be an arbitrary Noetherian ring. An object $\omega^\bullet\in \mathrm{D}(A)$ is a *dualizing complex* if

(1) $\omega^\bullet$ has finite injective dimension, i.e. there exists $\alpha\in\mathbb{Z}$ such that for every $X\in \mathrm{D}^{\heartsuit}(A)$,

$$\mathrm{RHom}_A(X,\omega^\bullet)\in \mathrm{D}^{\leq\alpha}(A);$$

this is also equivalent to that $\omega^\bullet$ is represented by a bounded complex of injectives.

(2) $\omega^\bullet\in \mathrm{D}_{\mathrm{coh}}(A)$.

(3) the natural map is an isomorphism:

$$A=\mathrm{RHom}_A(\omega^\bullet,\omega^\bullet).$$

Note that dualizing complexes are stable under localization (Stacks 0A7G).

Theorem 3.2. *If A is local, then*

$$\omega^\bullet\otimes^L_A\widehat{A}\simeq \omega^\bullet_{\widehat{A}}[\delta]$$

for some $\delta\in\mathbf{Z}$.

We say $\omega^\bullet$ is *normalized* if $\delta=0$.

Proof. It suffices to show (by Prop 2.3 and the fact that RHom commutes with flat base change if the source in pseudo-coherent)

$$\mathrm{RHom}_A(k,\omega^\bullet)\simeq k[\delta].$$

From conditions (1) and (3), for $M\in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(A)$,

$$M=\mathrm{RHom}_A(\mathrm{RHom}_A(M,\omega^\bullet),\omega^\bullet).$$

In particular,

$$k=\mathrm{RHom}_A(\mathrm{RHom}_A(k,\omega^\bullet),\omega^\bullet).$$

By (1) and (2), $\mathrm{RHom}_A(k,\omega^\bullet)\in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(k)$, hence is of the form $\bigoplus_{i=1}^t k[a_i]^{\oplus d_i}$. Then

$$k=\mathrm{RHom}_A(\mathrm{RHom}_A(k,\omega^\bullet),\omega^\bullet)\cong\bigoplus_{j=1}^t\left(\bigoplus_{i=1}^t k[a_i]^{\oplus d_i}\right)[a_j]^{\oplus d_j},.$$

forcing $t=1=d_1$; and we put $\delta=a_1$. □

Remark 3.3. Not all Noetherian rings have dualizing complexes. If a local ring has two, they are isomorphic up to a shift. This is because if $X, Y \in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(A)$ are such that $X \otimes^L_A \widehat{A} \simeq Y \otimes^L_A \widehat{A}$, then $X \simeq Y$. This is because $\mathrm{Hom}_{\mathrm{D}(A)}(X, Y)$ is dence in its completion and because

$$\{\text{quasi-isomorphisms}\} + \mathfrak{m}\,\mathrm{Hom}_{\mathrm{D}(\widehat{A})}(X \otimes^L_A \widehat{A}, Y \otimes^L_A \widehat{A}) \subseteq \{\text{quasi-isomorphisms}\}$$

by Nakayama's Lemma.

4. Properties of dualizing complexes

Lemma 4.1. *Let $A \to B$ be a finite ring map, and let $\omega^\bullet$ be a dualizing complex of A. Then*

$$\omega^\bullet_B := \mathrm{RHom}_A(B, \omega^\bullet)$$

is a dualizing complex of B.

If both A and B are local and $\omega^\bullet$ is normalized, then $\omega^\bullet_B$ is normalized.

Proof. For $N \in \mathrm{D}(B)$, the adjunction gives

$$\mathrm{RHom}_B(N, \mathrm{RHom}_A(B, \omega^\bullet)) = \mathrm{RHom}_A(N, \omega^\bullet).$$

Thus properties (1) and (2) in Def 3.1 follow. For property (3),

$$\mathrm{RHom}_B(\omega^\bullet_B, \omega^\bullet_B) = \mathrm{RHom}_A(\omega^\bullet_B, \omega^\bullet) = B,$$

by biduality for the finite A-module B.

For the normalized statement, we have

$$\mathrm{RHom}_B(l, \omega^\bullet_B) = \mathrm{RHom}_A(l, \omega^\bullet),$$

where l is the residue field of B, which is isomorphic to a finite direct sum of k as an A-module. □

5. Dimension function

For a normalized dualizing complex $\omega^\bullet$, define $\delta(\mathfrak{p})$ to be the unique integer d such that

$$\mathrm{H}^d(k(\mathfrak{p}), \omega^\bullet_{\mathfrak{p}}) \neq 0.$$

If A is local then $\omega^\bullet$ is normalized if and only if $\delta(\mathfrak{m}) = 0$.

Theorem 5.1. *If $\mathfrak{p} \subseteq \mathfrak{q}$ and $\mathrm{ht}(\mathfrak{q}/\mathfrak{p}) = 1$, then*

$$\delta(\mathfrak{p}) = \delta(\mathfrak{q}) + 1.$$

This means that δ is a dimension function.

Proof. By localization and quotient (Lemma 4.1) we may assume $(A, \mathfrak{m})$ is local, $\mathfrak{q} = \mathfrak{m}$, and $\mathfrak{p} = 0$, so $\dim A = 1$. We may also assume $\omega^\bullet$ is normalized. Then

$$\widehat{\mathrm{H}^{-1}(\omega^\bullet)} \simeq \mathrm{H}^1_{\mathfrak{m}}(A)^\vee, \qquad \widehat{\mathrm{H}^0(\omega^\bullet)} \simeq \mathrm{H}^0_{\mathfrak{m}}(A)^\vee.$$

Since $\mathrm{H}^0_{\mathfrak{m}}(A)$ is of finite length (being an ideal of A)[1], the finitely generated module $\mathrm{H}^0(\omega^\bullet)$ is also of finite length. Since $\mathrm{Supp}(\omega^\bullet) = \mathrm{Spec}\, A$ we must have $\mathrm{Supp}\, \mathrm{H}^{-1}(\omega^\bullet) = \mathrm{Spec}\, A$, whence by definition $\delta(\mathfrak{p}) = 1$. □

Corollary 5.2. *If a Noetherian ring A has a dualizing complex then A is catenary,* i.e.

$$\mathrm{ht}(\mathfrak{r}/\mathfrak{q}) + \mathrm{ht}(\mathfrak{q}/\mathfrak{p}) = \mathrm{ht}(\mathfrak{r}/\mathfrak{p})$$

for primes $\mathfrak{r} \supseteq \mathfrak{q} \supseteq \mathfrak{p}$.

[1] In fact, since we assumed A is a domain, we have $\mathrm{H}^0_{\mathfrak{m}}(A) = 0$.

6. Support of a cohomology module

Theorem 6.1. *Suppose A is local and let $\omega_A^\bullet$ be a normalized dualizing complex (if exists). Let $d = \dim A$. Then*

$$\operatorname{Ass}_A \mathrm{H}^{-d}(\omega_A^\bullet) = \{\mathfrak{p} \mid \dim A/\mathfrak{p} = d\}.$$

Proof. For every $\mathfrak{p} \in \operatorname{Spec} A$, $\operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet)$ is a normalized dualizing complex of $A/\mathfrak{p}$, hence is in $\mathrm{D}^{[-\dim A/\mathfrak{p}, 0]} \subseteq \mathrm{D}^{[-d,0]}$ (Lem 4.1). If there is an injection

$$A/\mathfrak{p} \hookrightarrow \mathrm{H}^{-d}(\omega_A^\bullet),$$

then

$$\mathrm{H}^{-d} \operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet) = \operatorname{Hom}_A(A/\mathfrak{p}, \mathrm{H}^{-d}(\omega_A^\bullet)) \neq 0.$$

It follows that $\dim A/\mathfrak{p} \geq d$.

Conversely, if $\dim A/\mathfrak{p} = d$, then $\delta(\mathfrak{p}) = -d$ (Th 5.1), that is,

$$\left(\mathrm{H}^{-d} \operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet)\right)_{\mathfrak{p}} \neq 0.$$

Hence there is a nonzero map

$$k(\mathfrak{p}) \longrightarrow \mathrm{H}^{-d}(\omega_A^\bullet)_{\mathfrak{p}},$$

which gives an injection $A/\mathfrak{p} \hookrightarrow \mathrm{H}^{-d}(\omega_A^\bullet)$ that localizes to the given map. □

Corollary 6.2.

$$\operatorname{Supp} \mathrm{H}^{-d}(\omega_A^\bullet) = \bigcup_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \dim A/\mathfrak{p} = d}} V(\mathfrak{p}).$$

The following statements are left as exercises.

(1) $$\operatorname{Ass}_A \mathrm{H}^{-j}(\omega_A^\bullet) \supseteq \{\mathfrak{p} \in \operatorname{Min} A \mid \dim A/\mathfrak{p} = j\}$$

(2) $$\dim \operatorname{Supp} \mathrm{H}^{-j}(\omega_A^\bullet) \leq j$$

Here $\operatorname{Min} A$ is the set of minimal primes of A.

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